Chulalongkorn University · Department of Chemical Engineering

Natural Draft Fired Heater Ver 1 · H-100 Operator Training Simulator

Relief load basis and relief-device sizing

NDFH-V1-PE-RLB-001 · Rev 0 · Issued for teaching

Rev	Date	Description	Originator	Checker	Approver
0	25-Oct-26	Issued for teaching	A. Soottitantawat	—	—

Title block	
Client	Chulalongkorn University · Department of Chemical Engineering
Project	Natural Draft Fired Heater Ver 1 · H-100 Operator Training Simulator
Document no.	NDFH-V1-PE-RLB-001
Title	Relief load basis and relief-device sizing
Drawings	PFD-100 rev 1; P&ID-001 rev C (screen drawing); NDFH-V1-PID-001 rev E (EPC alternative)
Scope	Overpressure scenarios of H-100, E-100 and the fuel oil train; relief loads; API 520 sizing; model relief runs.

All numbers in this document come from the Natural Draft Fired Heater Ver 1 simulator model (design case and model runs) or are stated as assumed. It is a teaching document, not a design for construction.

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1 · Basis

Scenarios after API 521 (blocked outlet, external fire, thermal expansion, tube rupture, control-valve failure, utility failure). Relieving pressure: set + 10 % accumulation for process cases, + 21 % for fire. Sizing after API 520 Part I (SI): vapour $A = W/(C\,K_d\,P_1\,K_b\,K_c)\,\sqrt{(T\,Z/M)}$, K_d 0.975; liquid $A = 11.78\,Q/(K_d\,K_w\,K_c\,K_v)\,\sqrt{(G/(P_1 - P_2))}$, K_d 0.65. Orifices from API 526.

2 · Relief devices

Tag	Protects	Set kPa(g)	Status	Governing case	Load	Area req. mm²	Orifice	Discharge
PSV-101	E-100 tube side (feed)	1700	P&ID	Fire (vapour)	3893 kg/h	158	F (198 mm²)	Closed drain
PSV-103	E-100 shell side (heating fluid)	1000	recommended	Tube rupture (liquid)	19495 kg/h	199	G (325 mm²)	Unit A hot-oil drain (EQ)
TSV-120	Fuel oil, XV-301 to EIV-120 (blocked in)	2200	recommended	Thermal expansion	—	—	¾ × 1 in	Back to the fuel tank
PSV-102	Coil outlet (or V-200 relief)	1700	recommended / confirm	Blocked outlet with firing	—	—	by study	V-200 relief system

3 · Relief load basis, case by case

Device	Case	Basis	Result
PSV-101	Blocked outlet (FV-010 or V-106 closed)	Pump shut-off 1450 kPa(g) < set 1700 kPa(g)	Not a relief case (pump cannot exceed the set pressure)
PSV-101	Thermal expansion, tubes blocked in, hot fluid flowing	$Q = \alpha_v\,\Phi/(\rho\,c_p) = 3.43$ m³/h (α_v 0.001 /K, Φ = 1904 kW)	A = 21.4 mm² (liquid)
PSV-101	External fire, liquid-full, adequate drainage	$Q = 43.2\,F\,A^{0.82} = 324$ kW (A = 11.7 m²); W = Q/λ = 3893 kg/h	A = 158 mm² (vapour, 121 % accumulation) → F orifice — governing
PSV-103	Tube rupture (2 × tube cross-section)	Shell design 1000 < 10/13 × 1700 = 1308 kPa(g) → case applies; W = 19495 kg/h feed at ΔP 430 kPa	A = 199 mm² (liquid, 110 %) → G orifice
PSV-102	Blocked outlet, burner firing	Coil at pump shut-off 1450 kPa(g) < 1700; vaporisation needs the burner to keep firing	FSLL-008 (SIL 2) stops firing; confirm in the relief study whether the SIF is credited (EQ)
TSV-120	Blocked-in fuel oil heated by sun or fire	Liquid trapped between EIV-120/V-121 and XV-301	Minimum size thermal relief ¾ × 1 in
Explosion doors	Firebox overpressure after delayed ignition	Casing design ±(vendor) mmH₂O	Doors on the radiant section (Safety layers tab, layer 4)

4 · Model relief runs

Each run starts from design on a copy of the model, applies one failure with no operator action, and runs 10 min with the trips disabled and again with the SIS active. Pressures come from the pump and fuel-supply relations of the model.

Failure	SIS	P max kPa(g)	Design	Skin max °C	Outlet max/min °C	λ min	CO max ppm	TI-010 max	Trip
FV-010 fails closed (feed blocked in)	disabled	1400	1700	675	406 / 326	1.11	20	180.0	none in 10 min
FV-010 fails closed (feed blocked in)	active	1400	1700	566	378 / 304	1.11	20	180.0	FSLL-008 low process flow at 9.2 s
FV-300 fails wide open (fuel)	disabled	1351	2200	398	350 / 313	0.74	5183	180.0	none in 10 min
FV-300 fails wide open (fuel)	active	1351	2200	398	350 / 313	0.74	5183	180.0	none in 10 min
HV-063 fails wide open (heating fluid)	disabled	1030	1700	402	360 / 350	1.11	20	220.5	none in 10 min
HV-063 fails wide open (heating fluid)	active	1030	1700	402	360 / 350	1.11	20	220.5	none in 10 min
FV-200 fails closed (combustion air)	disabled	—	—	398	350 / 179	0.58	2898	180.0	none in 10 min
FV-200 fails closed (combustion air)	active	—	—	398	350 / 175	0.58	2898	180.0	BSLL-005 flame failure at 6.5 s

Lessons: a blocked feed cannot overpressure the coil (pump shut-off below design), but with the trips disabled the tube skin reaches the value in the table; FSLL-008 holds it lower. A fuel valve failing open with the air unchanged makes the firing fuel-rich: the outlet falls while CO climbs.

Assumptions used in this document

No.	Assumed	Value	Used in
A1	Feed viscosity	1.0 mPa s at 40 °C, 0.5 mPa s at 180–350 °C (Design tab value)	Line sizing
A2	Feed vapour pressure at 40 °C	5 kPa(a)	NPSH available
A3	Pump efficiency P-100A/B	60 % at rated flow	Absorbed power, motor size
A4	Heating fluid from Unit A	hot oil, ρ 850 kg/m³, c_p 2.3 kJ/kg K, return 150 °C, operating 600 kPa(g), design 1000 kPa(g) / 260 °C	E-100 rating, tube-rupture case
A5	E-100 overall U	350 W/m² K (liquid–liquid, clean), 15 % area margin	E-100 area
A6	E-100 geometry	TEMA AES, ¾ in 14 BWG tubes (ID 14.8 mm), shell OD 0.61 m × 6.1 m	Wetted area, rupture area
A7	Fuel oil	ρ 900 kg/m³, μ 10 mPa s, delivered heated to 120 °C, flash point 66 °C	Fuel line, area classification
A8	Feed fluid	flash point 45 °C, auto-ignition 250 °C, latent heat 300 kJ/kg, vapour M 150, k 1.05 at relief	Relief and area classification
A9	Allowable stresses	illustrative ASME II-D values (see the stress table); confirm from the code edition in force	Wall thickness
A10	Electrical	400 V, 3-phase, 50 Hz; motor η 92 %, pf 0.86; starting current 6 × FLC	Load list, cable sizing
A11	Lead times	typical market lead times in weeks, for teaching	Procurement schedule
A12	Corrosion allowance	3 mm carbon steel coil and piping, 3 mm vessels	Wall thickness

5 · Checks

Check	This document	Compared with	Result
PSV-101 set ≤ E-100 tube design pressure	1700 kPa(g)	1700 kPa(g)	✓
PSV-101 does not lift in normal operation (set ≥ 1.1 × PI-002)	1700 kPa(g)	1133 kPa(g)	✓
Blocked outlet cannot reach PSV-101 set (pump shut-off)	1450 kPa(g)	1700 kPa(g)	✓
Tube-rupture case on E-100 shell: shell DP vs 10/13 of tube DP	1000 kPa(g)	1308 kPa(g) → case applies	info
PSV-101 orifice ≥ required area (Fire (vapour))	F = 198 mm²	158 mm² required	✓
PSV-103 orifice ≥ required area (Tube rupture (liquid))	G = 325 mm²	199 mm² required	✓
A coil relief device is fitted or its absence is justified	PSV-102 recommended	Safety layers tab: "confirm downstream relief covers it"	info
Run: FV-010 fails closed (feed blocked in) — highest pressure	1400 kPa(g)	1700 kPa(g) design	✓
Run: FV-300 fails wide open (fuel) — highest pressure	1351 kPa(g)	2200 kPa(g) design	✓
Run: HV-063 fails wide open (heating fluid) — highest pressure	1030 kPa(g)	1700 kPa(g) design	✓